

Pneumonia Detection from Chest X-Rays

Interim Business Report

RSNA Pneumonia Detection Challenge
Deep Learning · Convolutional Neural Networks · TensorFlow / Keras
Apple M1 GPU (TensorFlow Metal) · Python 3.12

Dataset	RSNA Pneumonia Detection Challenge (Kaggle)
Total Patients	26,684 unique chest X-rays (DICOM format)
Framework	TensorFlow 2.16 / Keras — Apple M1 GPU
Baseline Result	Scratch CNN — Test ROC-AUC 0.8353, pneumonia recall 0.84
Submission Type	Interim Business Report
Date	July 2026
Companion Notebook	src/pneumonia_interim.ipynb (all outputs reproduced here)

Interim Evaluation Rubric — Coverage Map

Rubric Section	Points	Covered In
Data Overview — import data, check shape	6	Section 3
EDA — samples per class, imbalance, observations	8	Section 4
Preprocessing — grayscale, before/after, split, normalization	10	Section 5
Model Building — CNN from scratch, training, performance	10	Section 6
Business Report Quality	6	Whole document
Total	40	

Per the business-report guidelines, every section below is organised into four parts: **What we did**, **Why we did it**, **What we observed**, and **What we concluded**. No source code is included — key outputs are reproduced as captured results and figures from the companion notebook.

1. Executive Summary

This interim report covers the first phase of building an automated pneumonia detection system from chest X-rays: data understanding, exploratory analysis, preprocessing, and a convolutional neural network (CNN) trained from scratch as a baseline.

Working from **26,684 DICOM chest X-rays** (RSNA Pneumonia Detection Challenge), we established that true pneumonia prevalence is **22.5%** — not the 31.6% the raw label file suggests — built leakage-safe stratified splits, and trained a 126,849-parameter CNN with class weighting. The baseline achieved **Test ROC-AUC 0.8353** and, importantly for a screening tool, **84% pneumonia recall** at the default decision threshold — the class-weighted loss shifted the model toward the clinically safer operating point.

These results validate the full pipeline and set a strong baseline for the final phase, where transfer learning with pretrained backbones, threshold governance, and deployment as a web application will complete the system.

2. Business Context & Objectives

Pneumonia is a leading cause of death worldwide, disproportionately affecting children under five and the elderly. Diagnosis relies on chest X-ray interpretation by radiologists — a scarce resource, particularly in rural and resource-constrained settings. Fatigue and high patient loads introduce diagnostic variability; delayed reads delay antibiotic treatment.

The business goal is a decision-support system that automatically flags likely pneumonia cases for priority radiologist review — increasing throughput and consistency without removing the clinician from the loop. For such a screening tool, **missing a pneumonia case (false negative) is far costlier than a false alarm**; this asymmetry drives every modelling decision in this report.

3. Data Overview

3.1 What we did

We imported the three data sources, checked their shapes, and audited the relationship between label rows, patients, and image files:

```
=== stage_2_train_labels.csv === Shape: (30,227 rows × 4 columns)
=== stage_2_detailed_class_info.csv === Shape: (30,227 rows × 2 columns)
=== stage_2_train_images.zip === 26,684 DICOM files (~3.5 GB)
```

3.2 Why we did it

Every downstream number — class balance, split sizes, evaluation metrics — depends on knowing exactly what one 'sample' is. A mismatch between 30,227 label rows and 26,684 image files signals duplication that must be resolved before any analysis, or prevalence statistics and train/test splits would be silently wrong.

3.3 What we observed

Metric	Value
Label CSV rows	30,227
Unique patients / DICOM files	26,684
Duplicate rows (multi-box pneumonia patients)	3,543
Pneumonia patients after deduplication	6,012 (22.5%)
Raw-CSV pneumonia share (misleading)	31.6%
Image properties	1024×1024, uint8, single-channel grayscale
Pixel maxima across samples	230–255 (scanner-dependent)

Pneumonia-positive patients appear on one row per annotated opacity region, so the raw CSV overstates pneumonia prevalence. Images are natively grayscale (X-ray physics — no colour information exists), and maximum pixel intensity varies by scanner.

3.4 What we concluded

- All analysis must happen at **patient level** — labels deduplicated to one row per patient before any split or statistic.
- True class imbalance is **3.44 : 1**, more severe than the raw file implies — accuracy alone cannot be the success metric.
- Scanner-dependent intensity ranges require **per-image normalization** rather than a single global scale.
- No RGB-to-grayscale conversion is needed; a channel dimension is added programmatically for the CNN.

4. Exploratory Data Analysis

4.1 What we did

We visualised random X-rays from each of the three source classes, quantified the class distribution at patient level, and compared pixel-intensity statistics across classes (100 sampled patients per class).

4.2 Why we did it

Before modelling, we need evidence that (a) the classes are visually distinguishable — otherwise no classifier can succeed; (b) the imbalance is quantified — it dictates loss weighting and metric choice; and (c) simple global statistics carry signal — a sanity check that the task is learnable from pixels.

4.3 What we observed

Class	Patients	% of Total
No Lung Opacity / Not Normal	11,821	44.3%
Normal	8,851	33.2%
Lung Opacity (Pneumonia)	6,012	22.5%

Sample Chest X-rays by Class (red box on Lung Opacity = estimated opacity region)

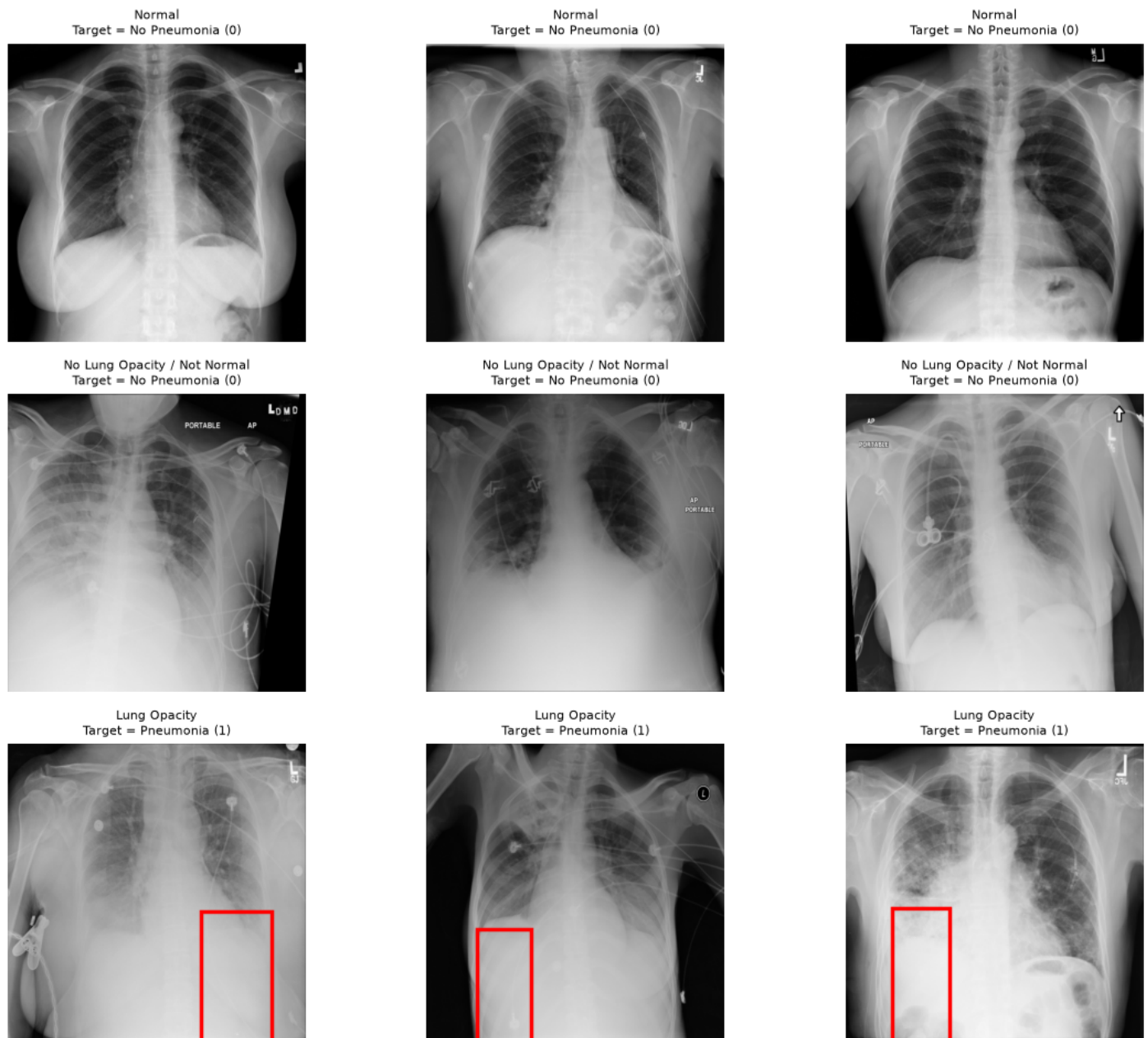


Figure 1 — Random samples per class. Normal: dark, air-filled lung fields with sharp costophrenic angles. Not Normal: real pathology (cardiomegaly, effusions) without pneumonia. Lung Opacity: focal white opacities in lung fields.

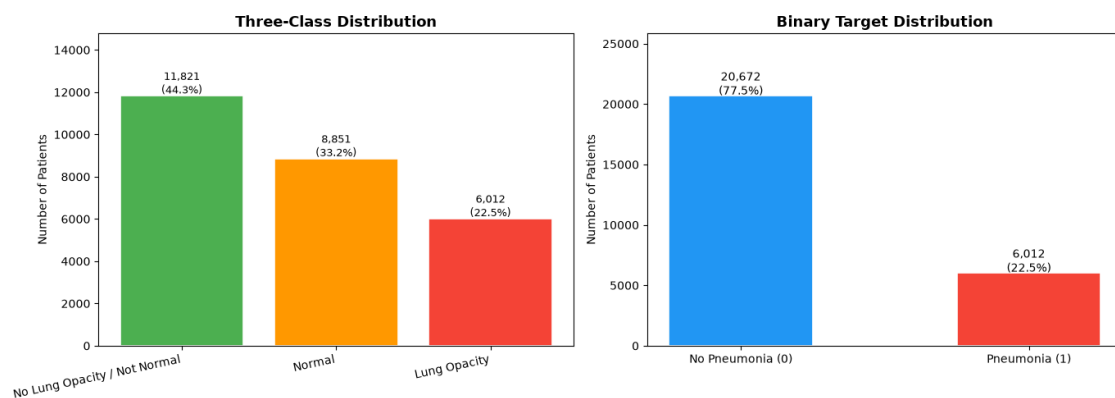


Figure 2 — Class distribution: binary imbalance is 77.5% vs 22.5% (3.44:1). A trivial always-negative classifier would score 77.5% accuracy while detecting zero pneumonia.

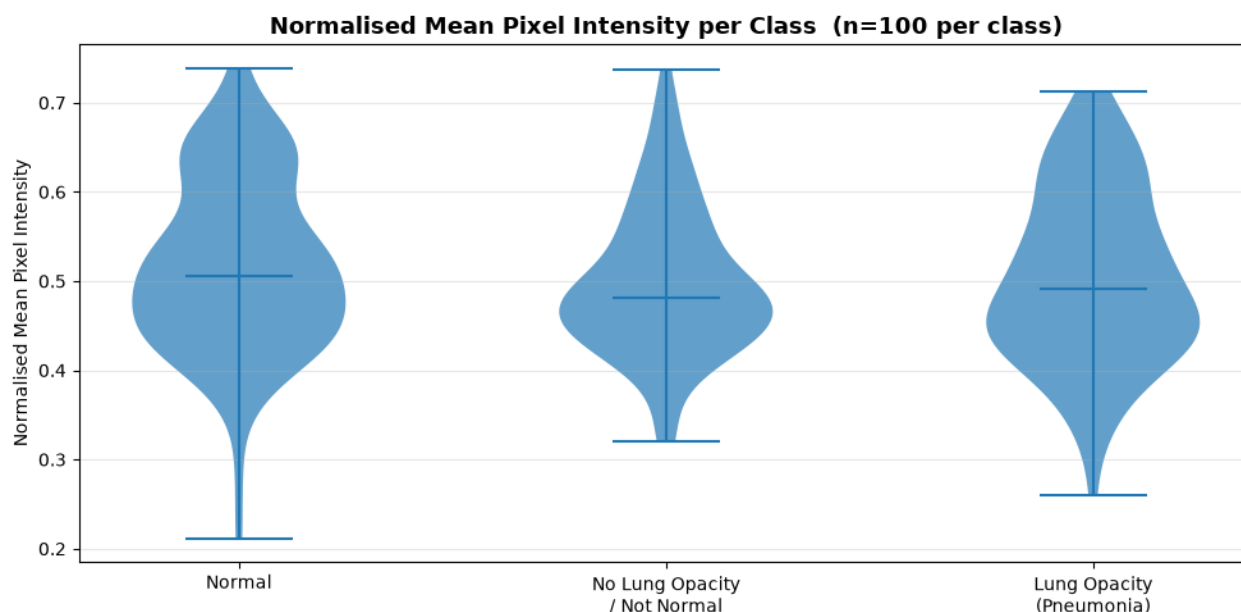


Figure 3 — Mean pixel intensity by class: pneumonia images are measurably brighter (opacities raise mean intensity); the 'Not Normal' class sits between Normal and pneumonia.

4.4 What we concluded

- Visual signatures exist and align with radiology: the task is learnable from pixels alone.
- The largest class (44.3%) is 'Not Normal' — pathology that mimics pneumonia — so false positives will concentrate there; the negative class is heterogeneous.
- Class weighting is required in the loss, and evaluation must lead with ROC-AUC and pneumonia recall, not accuracy.
- The intensity-separation finding motivates the CNN: if global brightness already separates classes weakly, learned spatial features should do far better.

5. Data Preprocessing

5.1 What we did

- Extracted 26,684 DICOMs to disk (one-time, ~3.5 GB) with a resume-safe guard.
- Confirmed images are natively grayscale; added the channel dimension (H, W) $\rightarrow$ (H, W, 1).
- Resized 1024×1024 $\rightarrow$ 128×128 and normalised each image by its own maximum to [0, 1].
- Split patients 70/15/15 (train/validation/test), stratified by the binary target.
- Defined an augmentation policy (random horizontal flip, small rotation, zoom) for training.

5.2 Why we did it

- **128×128** trades radiological detail for speed — appropriate for a baseline whose job is to validate the pipeline; pretrained backbones in the final phase use 224×224.
- **Per-image normalization** neutralises the scanner-dependent intensity ranges observed in Section 3 (maxima 230–255).
- **Patient-level, stratified splitting** guarantees no patient appears in two partitions (leakage) and that each partition keeps the 22.5% prevalence, making validation and test metrics directly comparable.
- **Augmentation** combats overfitting on 18,677 training images; chest X-rays are bilaterally symmetric, so horizontal flips are label-preserving.

5.3 What we observed

```
=== Dataset Split ===
```

```
Train : 18,677 patients (70%) | Pneumonia: 4,208 (22.5%)
```

```
Validation : 4,004 patients (15%) | Pneumonia: 902 (22.5%)
```

```
Test : 4,003 patients (15%) | Pneumonia: 902 (22.5%)
```

```
Patient-level split: no patient appears in both train and test.
```

Preprocessing: Before and After — One Sample per Class

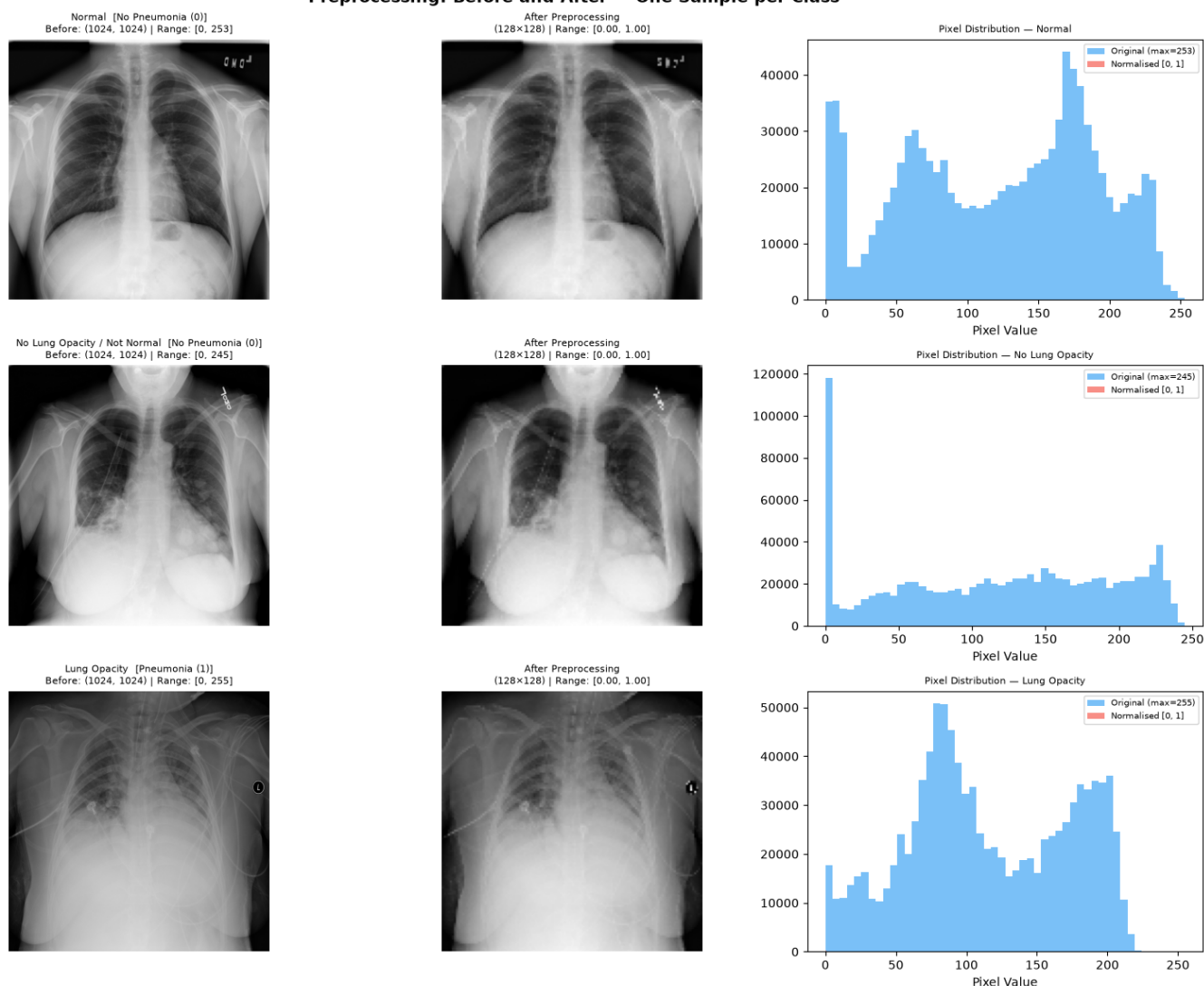


Figure 4 — Before/after preprocessing: the 1024×1024 original (left) retains its diagnostic structure after 128×128 resize and normalization (right).

Data Augmentation — Same X-ray, 5 Different Random Transformations

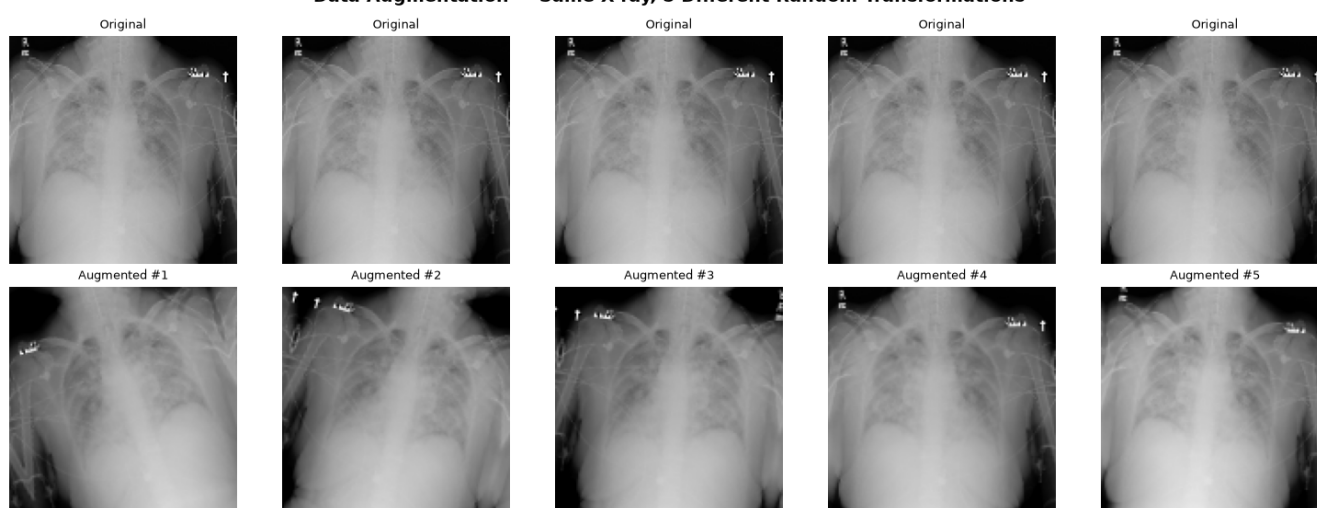


Figure 5 — Augmentation samples: each transform preserves the diagnostic label while varying presentation.

5.4 What we concluded

- Stratification worked exactly: 22.5% pneumonia in all three partitions — no distribution drift between validation and test.

- Normalization verified: all processed pixels lie in $[0, 1]$; non-zero minima reflect scanner baseline noise, handled correctly without clipping.
- Visual inspection (Figure 4) confirms opacities remain visible at 128×128 — the resolution is sufficient for a meaningful baseline.

6. Model Building — CNN from Scratch

6.1 What we did

Component	Choice	Rationale
Backbone	3 × (Conv2D → BatchNorm → MaxPool), filters 32/64/128	Progressive features: edges → textures → structures
Head	GlobalAveragePooling → Dense(256) → Dropout(0.5) → Sigmoid	GAP cuts parameters vs Flatten; Dropout fights overfitting
Size	126,849 parameters	Deliberately lightweight baseline
Loss	Binary cross-entropy with class weights {0: 0.645, 1: 2.219}	Each missed pneumonia costs ~3.4× more
Training	Adam 1e-3 · batch 32 · max 20 epochs · EarlyStopping + ReduceLROnPlateau on val AUC	Checkpoint keeps the best-validation model

6.2 Why we did it

A from-scratch CNN answers the question every stakeholder should ask before investing in larger models: **how much signal can a small, fast, fully-owned model extract?** It also establishes the honest baseline against which transfer learning must justify itself in the final phase. Class weighting encodes the clinical cost asymmetry directly into training rather than leaving it to after-the-fact threshold hacks.

6.3 What we observed — Training

Epoch	Val AUC	Note
1	0.7402	Initial convergence
2	0.7706	
4	0.6579	Dip — learning rate reduced to 5e-4
8	0.8060	Recovery after LR reduction
13	0.8208	
16	0.8266	
20	0.8357	★ Best — checkpoint saved

Validation AUC improved from 0.7402 to **0.8357** across 20 epochs (~110 s/epoch on Apple M1 GPU). The learning-rate schedule mattered: after the epoch-4 dip, ReduceLROnPlateau halved the rate and the model resumed steady improvement — augmentation kept the model from overfitting through all 20 epochs.

Training History

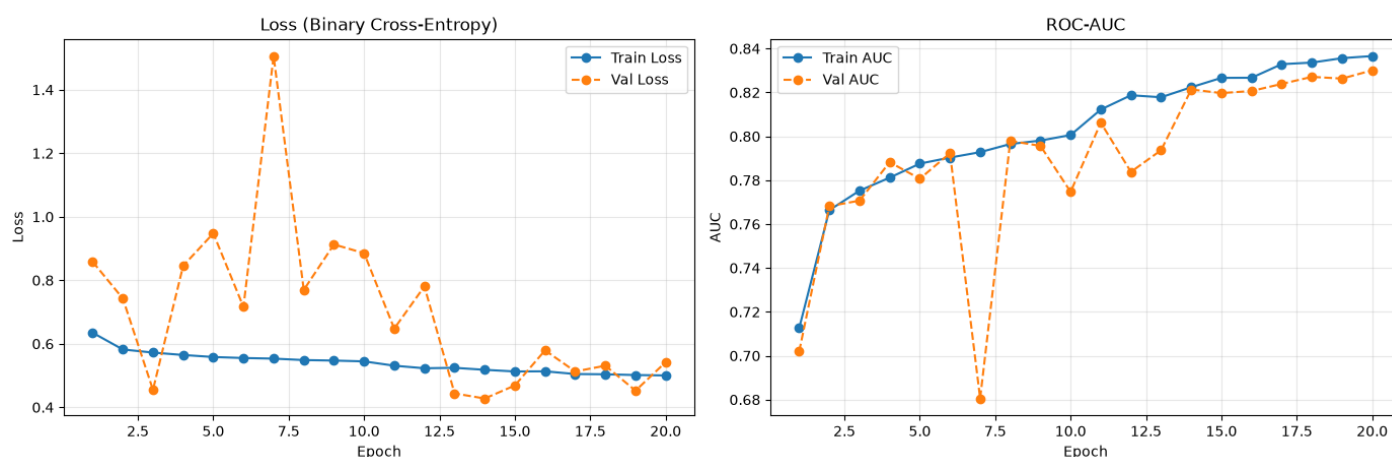


Figure 6 — Training curves: loss falling and AUC rising on both train and validation, with no divergence — a healthy fit.

6.4 What we observed — Test Set Evaluation

Test AUC : 0.8353 (4,003 patients: 902 pneumonia / 3,101 no pneumonia)

precision recall f1-score support

No Pneumonia (0) 0.93 0.65 0.77 3101

Pneumonia (1) 0.41 0.84 0.56 902

accuracy 0.70 4003

Test Set Evaluation | AUC = 0.8330

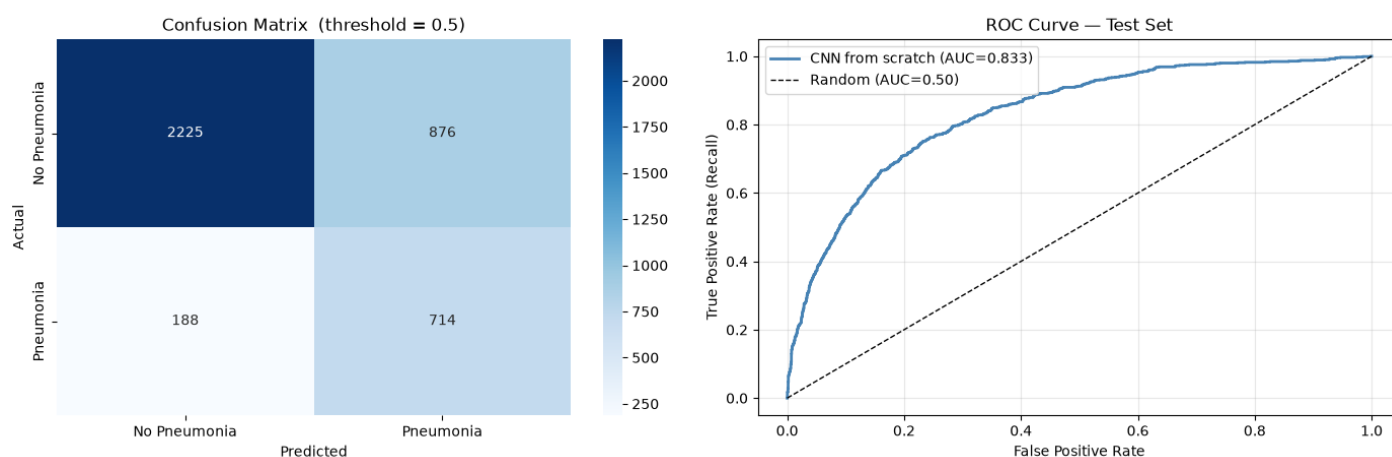


Figure 7 — Test evaluation: ROC curve (AUC 0.8353) and confusion matrix at threshold 0.5.

Precision-Recall Analysis

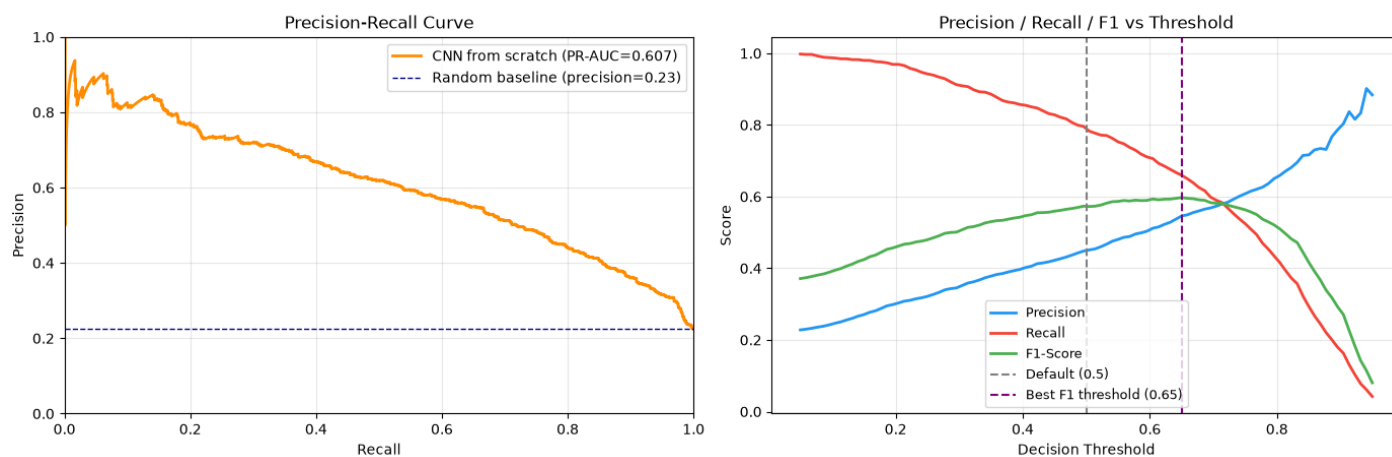


Figure 8 — Precision-recall curve and threshold sweep: the trade-off curve clinical stakeholders will use to set the operating point.

Most Confidently Misclassified Images

(red box = Grad-CAM: region model focused on for FN | orange box = region triggering false alarm for FP)

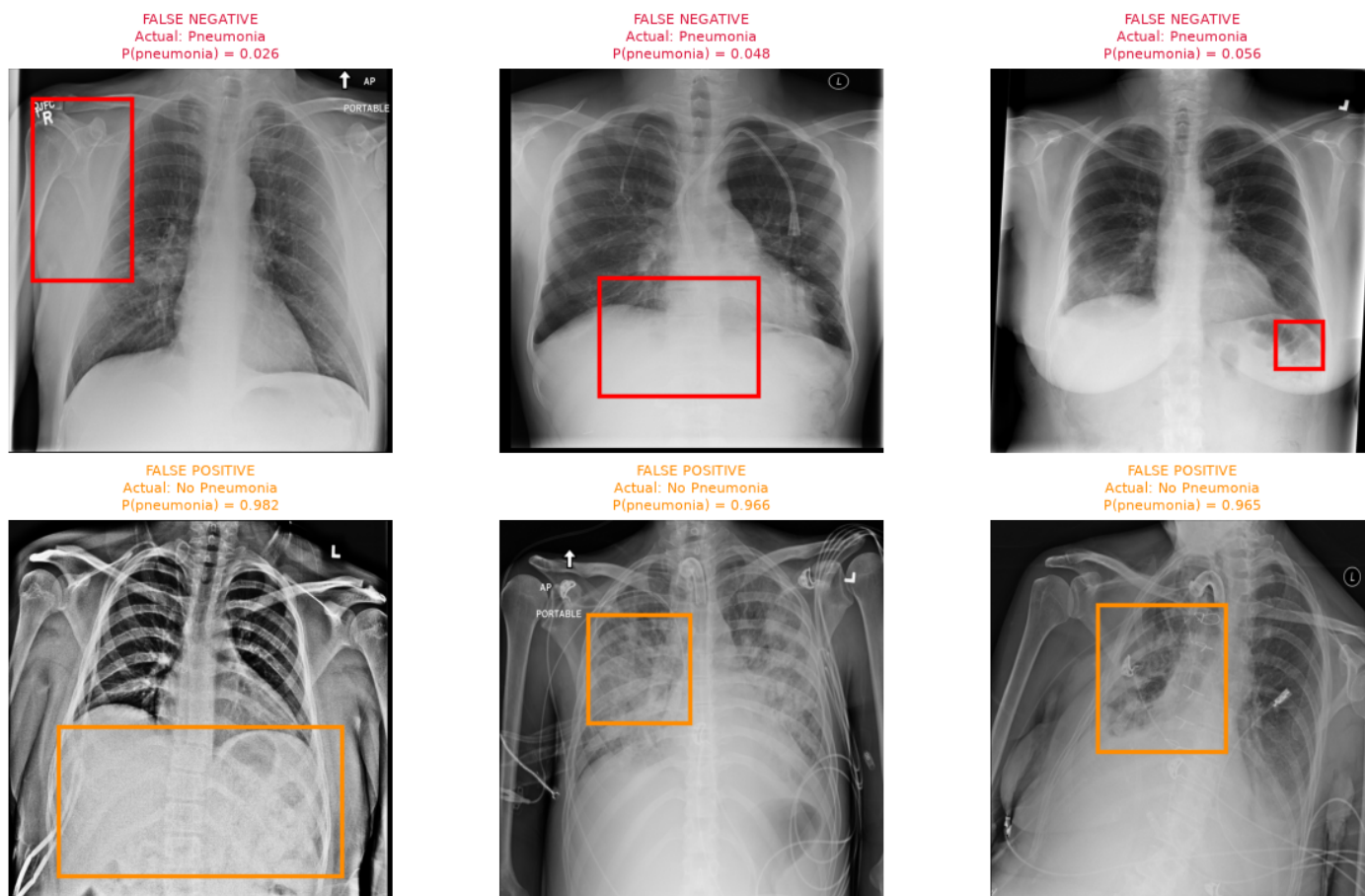


Figure 9 — Misclassification analysis with Grad-CAM heat-maps: false negatives are dominated by subtle, early-stage opacities; false positives come largely from the 'Not Normal' sub-class.

6.5 What we concluded

- **The baseline is clinically meaningful:** 84% of pneumonia cases are caught (758 of 902), with a highly trustworthy negative class (93% precision) — the profile a triage tool needs.
- **Class weighting worked as designed:** it moved the default operating point toward recall, rather than leaving the model to default to the majority class.
- **The cost is precision (0.41):** roughly 6 in 10 pneumonia flags are false alarms, mostly from the pneumonia-mimicking 'Not Normal' class — the clear target for improvement.
- **Headroom is visible:** AUC 0.8353 from a 126K-parameter model at 128×128 suggests pretrained backbones at 224×224 can push both AUC and precision higher — the central hypothesis for the final phase.

7. Overall Conclusions & Road Map

What this interim phase delivered:

- A verified, leakage-safe data pipeline from raw DICOM archives to model-ready tensors, with the true 22.5% prevalence established.
- Evidence-based preprocessing decisions (per-image normalization, patient-level stratified splits) — each justified by an observed property of the data.
- A scratch-CNN baseline at **Test ROC-AUC 0.8353 with 84% pneumonia recall** — already a usable screening profile, and an honest benchmark for what follows.

Road map for the final report:

Planned Work	Expected Business Outcome
Transfer learning: ResNet50V2 and EfficientNetB0, frozen then fine-tuned, at 224×224	Higher AUC and better precision at equal recall — fewer false alarms per detection
Validation-tuned decision thresholds (85% sensitivity target)	A defensible, documented operating point set on validation data, not guesswork
Model comparison and selection	A single best model chosen on clinical criteria, serialized for deployment
Streamlit web app in Docker + model registry	A reproducible, demonstrable tool any stakeholder can run and test

Companion notebook: `src/pneumonia_interim.ipynb` | Model checkpoint: `best_cnn_model.keras` | All figures and outputs in this report are reproduced from the executed notebook.